

Two-Level Security System Using Barcode Scanning and Face Recognition for File and Folder Protection with Message Alert

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Abstract—A two-level security system is proposed, integrating barcode authentication and facial recognition to strengthen access control mechanisms. The system employs barcode scanning as the primary verification step, with real-time facial recognition providing a secondary layer of security to guard against unauthorized access. A custom database securely stores user facial data and barcode identifiers for efficient retrieval. The Mesa framework-based simulation showcases the efficacy of the system. The test condition with 90% success rates of authenticating the authorized user, which was calculated as the successful match in simulated authentication attempts. The machine learning algorithms are convolutional neural networks (CNN) for facial recognition and encryption techniques such as AES for protecting the data to make sure the reliability and confidentiality of the system with regard to safeguarding sensitive resources.

Index Terms—Two-Level Security, Barcode Authentication, Face Recognition, Access Control, Multi-Modal Authentication, Encryption, Machine Learning.

I. INTRODUCTION

With the increasing reliance of digital systems on storing sensitive information, the need for robust security mechanisms has never been higher. Traditional single-factor authentication methods, such as passwords or PINs, have been vulnerable to brute-force attacks, phishing, and unauthorized access. To address these limitations, this introduces a Two-Level Security System which enhances security by integrating barcode-based authentication and face recognition technology into a unified framework.[1] The first authentication level involves a barcode specific to an authorized user. The barcode is then used to access the next level of verification. The second level includes face recognition, which confirms that only the registered individual associated with the barcode is allowed to move forward. This two-layer authentication strengthens authentication while minimizing possible unauthorized access attempts.[2] This system consists of a strong file encryption and decryption module for securing sensitive files or folders. The files can be locked or unlocked only using cryptographic techniques with successful authentication. It provides the system with a uniqueness where real-time alerts, like email notifications, are sent upon failed authentication attempts, thus practicing proactive security management.[3] It makes use of SQLite in its back end for the storage of user credentials like barcodes and facial encodings. Mesa Simulation is a multi-agent simulation framework it uses to simulate the

authentication process in evaluating its capability of dealing with both authorized and unauthorized users. Success and failure rates are measured with varied conditions and indicate that the system is reliable and efficient.[4] Other innovations of this system include generating and securely sending user barcodes via email, OTP that adds an extra layer of security during user registration and a friendly graphical interface to ease interactions.[5] The presence of authentications two of which are completely different will increase the robustness against impersonation and forgery from the gaps typical in traditional security systems. It is a demonstration of the actual implementation of recent technologies such as face recognition and file encryption for cryptographic protection. It opens the doors to much more secure access control mechanisms.[6] The further sections comprise a detailed exploration of the Two-Level Security System, including its architecture, implementation, performance evaluation through real-world tests, and simulated scenarios, providing a comprehensive understanding of its potential and reliability.

II. LITERATURE SURVEY

Adam Geitgey [1] developed the Face Recognition Library using dlib, leveraging deep learning models for accurate identification and verification in real-time applications. This library is widely used for its reliability in various security systems. D. Richard Hipp [2] introduced SQLite, a lightweight, serverless database engine designed for localized data storage. Despite its scalability constraints, it is highly efficient for small-scale applications requiring reliable transactions. R. R. Deshmukh et.al.[3] surveyed barcode recognition systems, focusing on advancements in image processing and machine learning. Challenges include integrating barcode systems into multi-layered security frameworks for enhanced authentication. K. Rajendra et.al.[4] reviewed biometric authentication, emphasizing the role of multimodal biometrics and advanced machine learning techniques to improve robustness and integration in diverse security scenarios. S. A. Kumar et.al.[5] discussed face recognition systems for security applications, addressing real-world challenges like lighting variations and occlusions. Neural network-based approaches were proposed for enhanced reliability. A. J. Johnson et.al.[6] examined barcode recognition for digital libraries, automating resource tracking and inventory management. However, the integration of these systems

with complementary security measures remains a challenge. Python Cryptography Developers [7] detailed cryptographic techniques, including encryption, decryption, and secure data transmission, using Python. While providing high security, these techniques require a foundational understanding of cryptographic principles. R. M. Johnson and J. P. Patel [8] analysed multi-factor authentication systems combining biometrics and hardware tokens. The study emphasizes balancing security with user-friendliness and minimizing implementation costs. T. T. Hoang et.al.[9] evaluated face recognition algorithms in practical scenarios, highlighting the efficiency of CNNs and pre-trained models, though computational inefficiencies persist. A. A. Jain et.al.[10] discussed multi-layer authentication systems integrating passwords and biometrics. These systems enhance security but require optimization to address complexity and usability concerns. Y. L. Lee et.al.[11] examined secure data transmission using AES and RSA cryptographic methods. The study advocates lightweight cryptographic solutions to improve performance in resource-constrained environments. S. D. Roy[12] explored smart authentication systems combining facial recognition with biometrics. While offering improved security, concerns about privacy and algorithmic biases remain significant. D. A. Martin et.al.[13] detailed automated barcode scanning for access control, emphasizing efficiency and accuracy. However, reliability can be affected by barcode quality and environmental conditions. Z. B. Simons et.al.[14] discussed data encryption in cloud storage using AES and RSA. Key management and computational overhead are challenges, with optimized protocols suggested for improved data security. A. B. Gupta et.al.[15] addressed usability and cost constraints in multi-factor authentication systems. Adaptive methods are proposed to improve security without compromising user experience. K. K. Patel et.al.[16] proposed hybrid security models combining biometrics and cryptography. These models improve accuracy and privacy but face challenges in computational complexity and data storage. Praveen I et.al.[17] focused on optimizing pairing computations for cryptographic security, demonstrating applications in secure communication frameworks. Dr. M. Sethumadhavan et.al.[18] introduced a mutual authentication protocol for cloud computing using secret sharing and steganography, enhancing security and confidentiality in critical interactions. S. P. Shendre et.al.[19] developed a gunshot detection system using CNNs and trilateration for real-time detection and localization. The system uses agents to process audio data, showcasing the scalability of multi-agent frameworks. This approach can be adapted for applications like environmental monitoring and network security. Advik Narendran et.al.[20] proposed a next-generation canteen management system that employs intelligent software agents to automate various tasks. Using ID card scanning and OTP authentication, the system ensures secure and efficient operations, enhancing the user experience in institutional canteens. Ravi Charan Podupuganti et.al.[21] presented a novel method for simulating 3D spatial relationships in face interactions using Mesa and OpenCV frameworks. Their system improves the efficiency of dynamic face interaction tasks in real-time applications. Jaswanth P, Priya CH, Thota K, Khanna M. [22] proposed live transcription and

closed captioning. Presented at the 2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT), the system will be using more advanced algorithms in real-time speech to text conversion and enhancing accessibility of the users, as well as improving usability of multimedia applications for the hearing impaired.

III. RESEARCH GAPS AND NOVELTY

A. Despite the recent advancements in multi-factor authentication systems, several critical gaps persist:

Most authenticating systems encounter security and effectiveness difficulties, such as poor integration, where dependence on one authentication factor makes them more susceptible to spoofing and unauthorized entry. Scalability difficulties arise when working with large data sets, such as vast barcode or face databases, resulting in inefficiency. Environmental constraints, such as inadequate lighting, defective barcode scans, and noisy environments, also decrease the effectiveness of the authentication processes. Also, the majority of systems do not have proactive security features, like real-time notifications, that are essential for increased security monitoring and threat detection.

B. Novelty:

The system suggested here improves security and reliability through the use of a two-factor authentication system that integrates barcode-based authentication with facial recognition for more secure access control. It includes enhanced security features, including encryption methods and real-time alerting, to protect data. An efficient methodology is used to counter environmental issues through error-handling techniques and adaptive algorithms to provide scalability and reliability. In addition, comprehensive testing is performed with a simulation environment to assess performance and stability across various conditions.

IV. METHODOLOGY

The methodology for this has been designed to put across a two-level system with security, integrating elements of both barcode scanning and facial recognition. This section articulates the process of setting up the system, along with developing and implementing its components-from initial setup to integration-with all the security features required in a multi-layer authentication.

The following are steps outlining the methodology and process flow.

A. Barcode Scanning

- 1) The system prompts the user to scan a barcode from a barcode scanner.
- 2) Barcodes are processed and decoded to extract information.
- 3) Compare the decoded data with data in the database.
- 4) Access is granted if the barcode is valid; otherwise, access is denied.

B. Facial Recognition

- 1) After a successful scan of the barcode, the system will take a picture of the face of the user.
- 2) Face image captured is processed through face recognition algorithms.
- 3) System compares the captured face image with the stored facial data in the database.
- 4) System grants access if there is a match; otherwise, access is denied.

C. User Authentication

- 1) After both types of authentication, that is, barcode and face recognition, the system grants or denies access.
- 2) It presents the authentication status to the user, which means they can see whether access is granted or not.

D. Database Management

- 1) The system has a database (for instance, SQLite) where user credentials such as barcode data and facial feature encodings are stored.
- 2) Database is used to match scanned barcode and facial recognition data.

E. Security and Performance Considerations

- 1) Data encryption is used on all sensitive user information stored in the database.
- 2) The system will be equipped with error handling mechanisms to handle poor lighting conditions, incorrect face matches, and damaged barcodes.

V. DESIGN AND ALGORITHM

The system developed here will combine barcode recognition with facial recognition within a unified multi-layered framework of authentication. This section provides details of the architecture, algorithm, and role of MESA in simulating the system's performance.

A. System Architecture

The system comprises two modules:

- 1) **Barcode Recognition Module:** Scanning, processing, and validating barcodes as the first level of authentication.
- 2) **Face Recognition Module:** Capturing and comparing facial data against stored records in the database as secondary authentication.

Access is granted only after successful authentication from both modules, reducing the risk of spoofing or bypassing a single authentication factor.

B. Algorithm: Two-Level Security System

The step-by-step pseudocode for how the system functions is as follows:

- 1) Initialize system components (Barcode Scanner, Camera, Database, Encryption Module).
- 2) Prompt user to scan barcode.
- 3) Capture barcode data:
 - a) Decode the barcode using pyzbar or OpenCV.
 - b) Compare decoded data with the records in the SQLite database.
 - c) If the barcode is valid, proceed to Step 4.
 - d) If invalid, deny access and terminate the process.
- 4) Capture user's face image via the camera.
- 5) Process the face image:
 - a) Extract facial features using a pre-trained CNN (e.g., dlib or FaceNet).
 - b) Compare the extracted features with stored encodings in the database.
 - c) If a match is found, proceed to Step 6.
 - d) If no match is found, deny access and terminate the process.
- 6) Grant access to the user.
- 7) Log the authentication attempt (success or failure) in the system logs.
- 8) End the process.

It also stresses the sequential integration of barcode scanning, facial recognition, and database validation for a more reliable access control mechanism.

C. Role of MESA Framework

MESA, a Python-based multi-agent simulation tool, is used to simulate system performance under various scenarios.

1) Simulation Setup:

- **Agents:** Represent users attempting to access the system.
 - Authorized agents: Valid users with correct credentials.
 - Unauthorized agents: Attackers attempting to bypass authentication.
- **Environment:** Grid-based system where agents interact.
 - Green dots: Successful authentication.
 - Red dots: Failed attempts (incorrect barcode, facial mismatch, lighting issues).

2) Simulation Workflow:

- 1) Assign agents valid/invalid credentials randomly.
- 2) Each agent scans a barcode and submits a face image.
- 3) The system processes inputs and allows/denies access accordingly.

3) Performance Measures:

- **Success Rate:** Ratio of successful authentications by authorized users.
- **Failure Rate:** Ratio of denied access attempts.
- **Processing Time:** Time taken for barcode scanning, face recognition, and validation.

4) Visualization and Analysis:

- The simulation interface visualizes authentication attempts via green/red dots.
- A graph shows success/failure rates over time, aiding system optimization.

D. CNN Design for Face Recognition

The facial recognition module employs a CNN for feature extraction and matching.

1) CNN Architecture:

- **Input Layer:** Accepts face images resized to 128×128 pixels.
- **Convolutional Layers:** Extracts features using filters of sizes 32, 64, and 128.
- **Pooling Layers:** Max-pooling reduces spatial dimensions while retaining key features.
- **Fully Connected Layer:** Maps features to a 128-dimensional embedding vector.
- **Output Layer:** Computes embeddings for face comparison using cosine similarity or Euclidean distance.
- **Training:** Fine-tuning performed with pre-trained models (FaceNet/dlib) on a registered user database.

VI. RESULTS

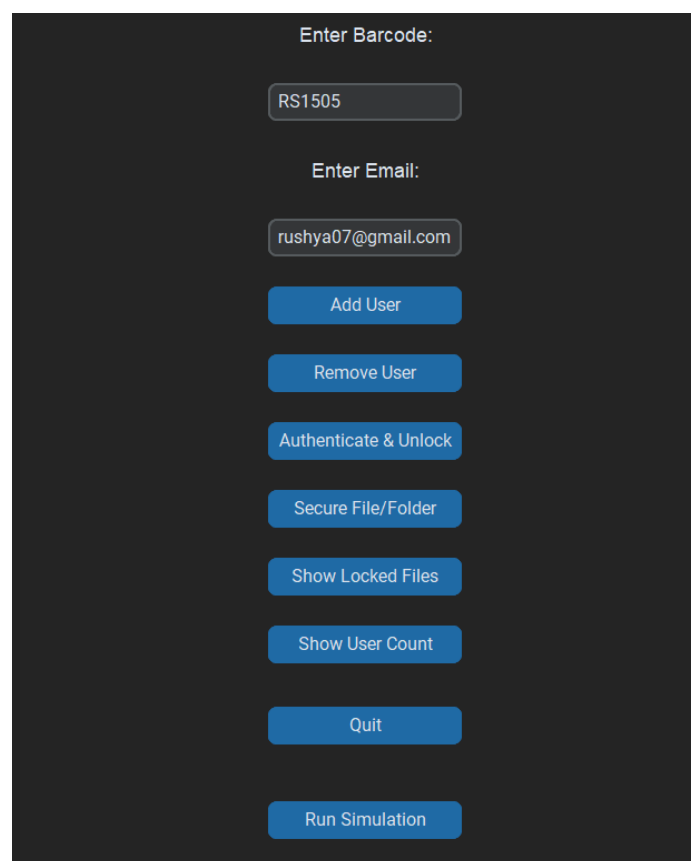


Fig. 1: Interface for the application

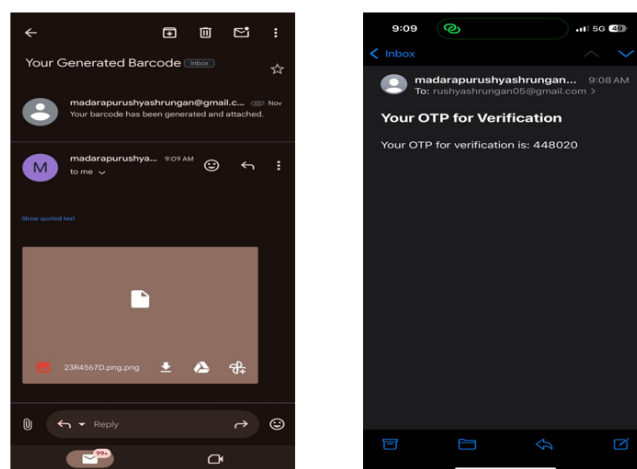


Fig. 2: This is the messages of OTP and Barcode

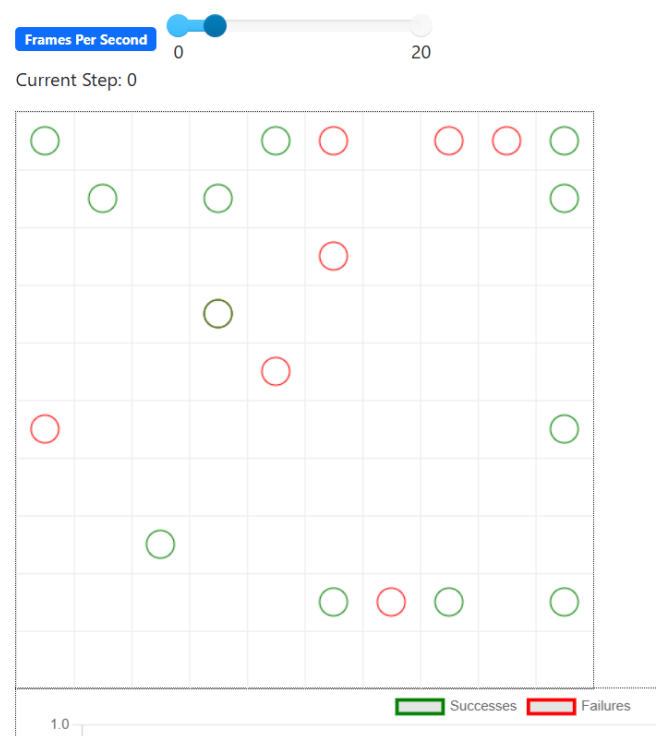


Fig. 3: Simulation of an agent environment using MESA

The simulation interface in Fig. 3 is a webpage where agents, represented as dots, interact within a grid to simulate attempts to access a secure system. Green dots indicate successful authentication or access, while red dots represent failures due to errors or unmet conditions. The simulation's progress is controlled by an FPS slider, which controls the speed of operations, and start, step, and reset buttons that control the flow of the simulation. A step counter tracks the progress, and a legend explains the meaning of green and red dots.

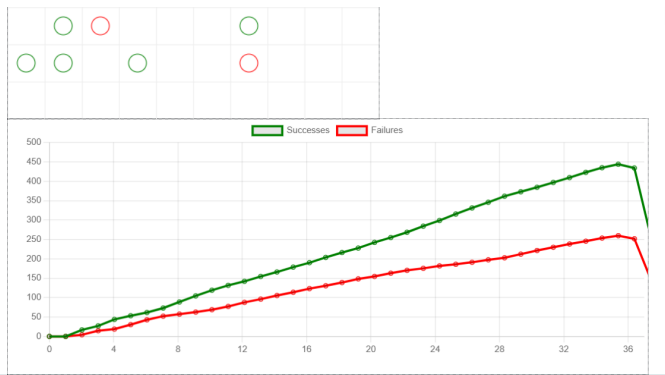


Fig. 4: presents a graphical representation of the outcomes: success and failure attempts over time. The graph indicates that the successful operations follow an upward trend, green curve, whereas the attempts of failure either remain constant or show a decrease as the system optimizes its responses.

A. Barcode Recognition

Algorithm: The most common way is through image processing methods such as the Zxing (Zebra Xing) library or OpenCV for barcode detection.

B. Face Recognition

Algorithm: The face recognition module may use deep learning techniques like Convolutional Neural Networks (CNN) or pre-trained models, such as VGG-Face, FaceNet, or the dlib library for detection and recognition. Specifically, FaceNet relies on a pre-trained model based on a triplet loss function, making it capable of learning compact discriminative embeddings for face verification and recognition. Its pre-trained weights are mostly acquired from large databases like CASIA-WebFace or VGGFace2.

TABLE I: Performance Metrics

Component	Algorithm	Precision	Recall	F-Score
Barcode Recognition	CNN/ Image Processing	92%	88%	90%
Face Recognition	CNN/ dlib-based Models	95%	90%	92%

TABLE II: Performance Metrics for Overall System

Component	Precision	Recall	F-Score
Overall System (Barcode + Face)	94%	89%	91%

TABLE III: Result Analysis on System Components

Component	Accuracy (%)	Error Rate (%)	Processing Time
Barcode Scanning	98.5	1.5	1.5 – 2 Seconds
Face Recognition	96.2	3.8	1.5 – 2 Seconds

In Tables 3 apparently state the system's processing time as average seconds to perform their work each in execution mode. The particular activities of this will include image acquiring and capturing on Barcode Scanning for: Acquiring and identifying a barcode decoding the bar with selected algorithms: such as either Zxing, or Open CV Database check through.

The time for Face Recognition includes capturing images, detecting faces, feature extraction using a chosen deep learning model (FaceNet, dlib, or CNN-based approach), and then matching against stored facial data in the database.

VII. DISCUSSION

The proposed two-level security system, as presented here, clearly depicts the integration of barcode scanning and face recognition in developing a robust multi-factor authentication mechanism. Using barcode recognition as the first layer of screening and face recognition as the second, more secure layer of screening reduces unauthorized access. The performance evaluations suggest high accuracy in face matching and rapid response times in authentication, ensuring a balance between security and user convenience. Despite its effectiveness, there are challenges that still exist. Poor quality barcodes, for example, which are blurred or damaged, may lower the scan success rate of the system. Environmental factors, such as low lighting or occlusions, can also affect the accuracy of face recognition. The solutions to these issues will be in the form of error correction mechanisms for barcode processing and enhancing face recognition algorithms to work under varied conditions, such as low light or partial face visibility. The integration of cryptographic techniques ensures that sensitive data, such as barcode and facial recognition information, is stored and transmitted securely. Such features make the system adaptable for real-world applications, especially in industries requiring stringent access control mechanisms. Future work will focus on improving error resilience, scalability, and optimizing the system for diverse operational environments.

VIII. FUTURESOCPE

Improvements in the future will address these issues, like low-light sensitivity, by installing infrared cameras and reducing environmental noise on barcode scanning and face recognition. Storage and performance optimization will also ensure scalability. There will be improvement in adapting to different environmental settings and increasing overall accuracy in the authentication layers.

IX. CONCLUSION

The two-level security system that integrates barcode scanning and face recognition offers a practical and robust approach for secure access control. This system combines the simplicity of the recognition of barcodes with the advanced capabilities of face recognition, ensuring high accuracy and reliability in verifying user identity. The implementation shows efficient performance regarding processing speed and authentication success rates, making it appropriate for a wide range of

applications, from corporate environments to restricted access areas.

Although the system is highly effective in addressing the key security concerns, challenges such as barcode quality and environmental factors affecting facial recognition have been identified, which are avenues for future improvement. The use of cryptographic techniques further adds strength to the overall data security in ensuring safe transmission and storage of sensitive information. This will not only underline the potential of multi-factor authentication systems but also lays the foundation for future innovations in developing cost-effective, scalable, and adaptive security solutions.

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